

7-4

Orthogonal Matrix

- Q is an orthogonal matrix
- $QQ^T = I_n$
- Q is invertible, and $Q^{-1} = Q^T$
- $Qu \cdot Qv = u \cdot v$ for any u and v
- $\|Qu\| = \|u\|$ for any u

Those properties are used to check orthogonal matrix.

44. Let W be a subspace of R^n . Let T be the linear operator on R^n defined by $T(v) = w - z$, where $v = w + z$, w is in W , and z is in $W^\perp$.

goal: $\|T(v)\|^2 = \|v\|^2$

(a) Prove that T is an orthogonal operator.

$$\|T(v)\|^2 = \|w - z\|^2$$

$$= (w - z)^T (w - z)$$

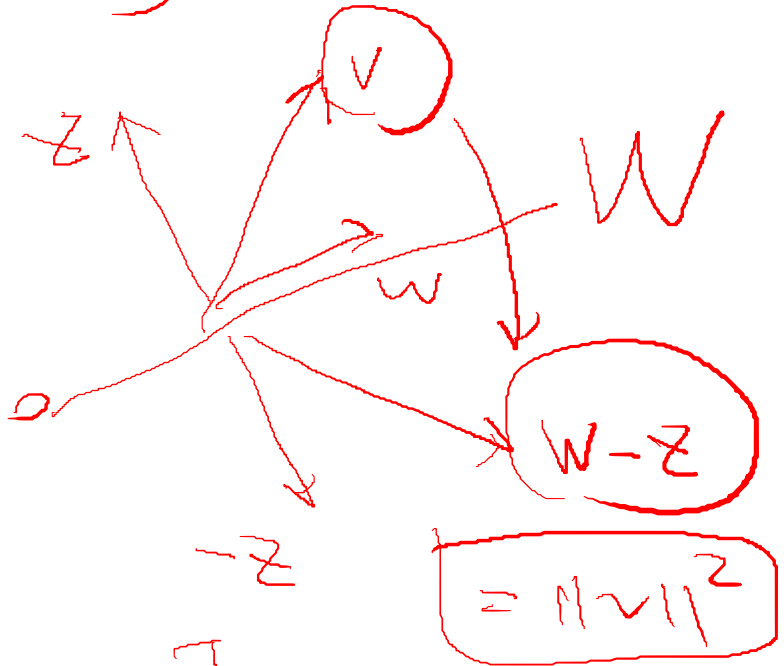
$$= w^T w - \cancel{z^T w} - \cancel{w^T z} + z^T z$$

$= 0 \quad = 0$

$$= w^T w + z^T w + w^T z + z^T z = (w + z)^T (w + z) = \|w + z\|^2$$

$$w \cdot z = 0$$

$$w^T z = z^T w = 0$$



(b) Let $U: R^n \rightarrow R^n$ be the function defined by $U(v) = \frac{1}{2}(v + T(v))$ for v in R^n . Prove that the standard matrix of U is P_W , the orthogonal projection matrix for W .

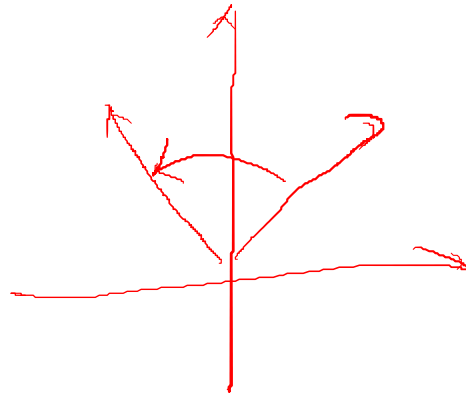
$$U(v) = \frac{1}{2}((w + z) + (w - z)) = w$$

49. Prove that if Q is an orthogonal matrix and λ is a (real) eigenvalue of Q , then $\lambda = \pm 1$.

$$Qv = \lambda v$$

$$\|\lambda v\| = \|v\|$$

$$\lambda = \pm 1$$



rotation:

orthogonal
eigen: no!

53. Let W be a 1-dimensional subspace of R^2 . Regard W as a line containing the origin. Let P_W be the orthogonal projection matrix on W , and let $Q_W = 2P_W - I_2$. Prove the following results:

(a) $Q_W^T = Q_W$ $Q_W Q_W = I$

(b) $Q_W^2 = I_2$ $Q_W Q_W^T = I$
 $||$

(c) Q_W is an orthogonal matrix.

$W = \text{Span}\{u\}, \|u\| = 1$

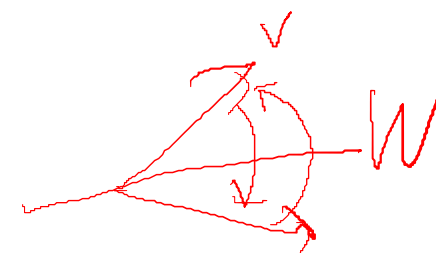
$P_W = C(C^T C)^{-1} C^T$ $C = \begin{bmatrix} u \end{bmatrix}$

$= u (u^T u)^{-1} u^T$

$= u u^T = I$

$Q_W = 2u u^T - I$

$(Q_W)^T = ?$ $Q_W Q_W = I$



$Q_W^2 = (2u u^T - I)(2u u^T - I)$
 $= 4u u^T - 2u u^T - 2u u^T + I$
 $= I$

53. Let W be a 1-dimensional subspace of R^2 . Regard W as a line containing the origin. Let P_W be the orthogonal projection matrix on W , and let $Q_W = 2P_W - I_2$. Prove the following results:

(d) $Q_W \mathbf{w} = \mathbf{w}$ for all $\mathbf{w}$ in W

$$Q_W \mathbf{w} = (2P_W - I) \mathbf{w}$$

$$= \frac{2P_W \mathbf{w}}{2\mathbf{w}} - \frac{I\mathbf{w}}{\mathbf{w}} = \mathbf{w}$$

$$P_W \mathbf{w} = \mathbf{w}$$

$$\forall \mathbf{w} \in W$$

(e) $Q_W \mathbf{v} = -\mathbf{v}$ for all $\mathbf{v}$ in $W^\perp$

$$Q_W \mathbf{v} = \frac{2P_W \mathbf{v}}{0} - \mathbf{v} = -\mathbf{v}$$

$$P_W \mathbf{v} = 0$$

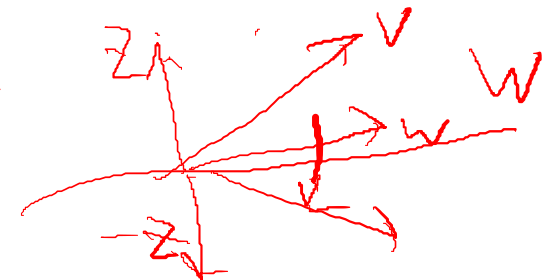
(f) Q_W is the standard matrix of the reflection of R^2 about W .

$$\forall \mathbf{v} \in R^2 \quad Q_W \mathbf{v} = (2P_W - I) \mathbf{v} = 2P_W \mathbf{v} - \mathbf{v}$$

$$P_W \mathbf{v} = \mathbf{w}$$

$$\mathbf{v} = \mathbf{w} + \mathbf{z}$$

$$= 2\mathbf{w} - (\mathbf{w} + \mathbf{z}) = \mathbf{w} - \mathbf{z}$$



54. Let T be a linear operator on R^n , and suppose that $\{v_1, v_2, \dots, v_n\}$ is an **orthonormal** basis for R^n . Prove that T is an orthogonal operator if and only if $\{T(v_1), T(v_2), \dots, T(v_n)\}$ is also an **orthonormal** basis for R^n .

$$\|v\|^2 = c_1^2 + c_2^2 + \dots + c_n^2$$

ind.

$$v \in R^n, v = c_1 v_1 + c_2 v_2 + \dots + c_n v_n, \|v\| = 1, v_i \cdot v_j = 0$$

$$\{ \|T(v_i)\| = 1, T(v_i) \cdot T(v_j) = 0 \}$$

$\Rightarrow$: basis? number, ind.

orthonormal?

$$\|T(v_i)\| = 1 \quad \checkmark$$

$$= \|v_i\| = 1$$

$$T(v_i) \cdot T(v_j) = 0?$$

$$= v_i \cdot v_j = 0 \quad \checkmark$$

$$\Leftarrow: T(v) = c_1 T(v_1) + c_2 T(v_2) + \dots + c_n T(v_n)$$

$$\|T(v)\|^2 = \|c_1 T(v_1) + c_2 T(v_2) + \dots + c_n T(v_n)\|^2 \Rightarrow \|T(v)\| = \|v\|$$

$$= c_1^2 + c_2^2 + \dots + c_n^2 = \|v\|^2$$

$\Rightarrow T$ is norm pre

55. Suppose that $\{v_1, v_2, \dots, v_n\}$ and $\{w_1, w_2, \dots, w_n\}$ are **orthonormal** bases for R^n . Prove that there exists a unique orthogonal operator T on R^n such that $T(v_i) = w_i$ for $1 \leq i \leq n$.

A

$$T(v_1) = w_1 \quad Av_1 = w_1$$

$$T(v_2) = w_2 \quad Av_2 = w_2$$

$\vdots$

$$T(v_n) = w_n \quad Av_n = w_n$$

$$A[v_1 \ v_2 \ \dots \ v_n] = [w_1 \ w_2 \ \dots \ w_n]$$

V

W

orthogonal matrix

$$QQ^T = I$$

$$\Rightarrow A = WV^{-1}$$

$$\begin{aligned} AA^T &= WV^{-1}(WV^{-1})^T = WV^{-1}(V^{-1})^T W^T \\ &= W \cancel{V(V^{-1})^T} W^T = WW^T = I \end{aligned}$$